Does Urbanisation Reduce Gender Inequality?

A Religion-Sensitive Statistical Analysis of Digital Access in India

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Abstract

The growth of the digital infrastructure in India is often linked with assumptions that urbanisation is a leveling factor, that may eliminate the old social inequalities including gender based inequality. This is the assumption in the majority of modern policy initiatives that have introduced urban digital infrastructure as a means of inclusion. The use of technology, however, is not an issue of physical access only, but it is an issue that is intensely refracted by social norms, cultural practices and institutional structures. In this research, the researcher will critically analyse the question of whether urbanisation lowers gender inequality in access to digital facilities across the religious factions in India.

Based on unit level data provided by the NSSO 80 th Round Modular Telecom Survey, the analysis will employ an intersectional approach combining gender, religion and sector (rural-urban). The research uses both cross tabulations, descriptive statistics, and composite indices to analyse access to digital devices, functional digital use as well as digital exclusion. Instead of considering the digital access as binary consequence, the paper views the qualitative aspects of the digital participation in terms of a Functional Digital Use Score and Digital Exclusion Score.

The results prove that urbanisation is a major contributor to the overall access to digital devices, but it does not eradicate gender inequalities. Even in urban regions women are still disproportionately locked out of digital access, and the degree of this lock out depends on religious communities. These findings indicate that, in India, digital inequality is socially, not spatially, organized. The paper has ended by stating that gender and religion sensitive policies of digital inclusion that focus on both social barriers and infrastructural shortfalls are required.

1. Introduction

In the modern social and economic life, the digital technologies have taken the central stage and influenced the access to the information, labor opportunities, access to the governmental services, education, and socialization. The state led processes like Digital India and the rapid growth in mobile connectivity and internet penetration in India has been seen to be accompanied by achievements through the use of technology to achieve inclusion in development. Cities, especially, are placed as the centres of digital transformation because of increased population density, enhanced infrastructure, and increased market incentives towards making a private investment.

Many of these policy discourses assume that urbanisation as a process works as an equaliser in itself. The city environment is usually depicted to be more liberal, less restricted by the traditional rules, as well as more accommodating of empowering women. In this story, the idea of the gender disparity in digital access is likely to reduce itself as the rural population will shift to the urban and become exposed to the modern infrastructure.

Nonetheless, an increasing literature refutes this supposition by showing that access to infrastructures does not always lead to fair utilization and advantage. Even in places where devices and connectivity exist, gender differences in technology usage exist, which would indicate that social norms and power dynamics still determine digital participation. Religion, as one of the most important identifiers of social identity in India, is critical in organising gender roles, mobility and engaging in public activities that are acceptable. These religiously knowledgeable norms can mediate through which women can experience and enjoy urban digital spaces.

This study addresses the following central research question:

Does urbanisation reduce gender inequality in digital access uniformly across religious groups in India?

The paper has gone beyond the aggregate rural-urban comparisons and has explored the variability of gender disparities in digital access according to religious groups in urban areas by adopting a religious sensitive approach. The research has three ways of contributing to the existing literature. It initially gives an in-depth statistical analysis of the difference in access to digital between genders based on nationally representative data. Second, it includes qualitative aspects of digital participation of functional use and measures of exclusion. Third, it places digital results of an individual level in the context of macro social and cultural determinations and connects micro level behaviour with macro level processes of urbanisation and social stratification.

2. Conceptual Framework and Literature Review

2.1 The Digital Divide: From Access to Use

Digital divide is a concept that has been changing over time. Initial debates were more concerned with differences in physical access to the computer and the internet, in many cases being a binary opposition between people with access and people without. Although access is a necessary requirement in digital participation, later studies have highlighted that it is not enough in comprehending digital inequality.

Modern scholarly literature differentiates several degrees of the digital divide. The former level relates to the availability of device and connection. The second tier is concerned with the disparities in the skills, the ways of its usage, and the digital literacy. The third tier looks at inequality in the results that include economic returns, social capital, and political participation based on digital engagement. This multi-dimensional approach helps emphasize the fact that people with equal access can have strikingly different digital consequences.

2.2 Gender and Technology

Gender has continuously become one of the major dimensions of digital inequality. Women tend to have low chances of owning or controlling digital devices compared to the men, using the internet in an independent manner, or doing functional digital tasks like online banking, e governance, and even employment related tasks. The causes of these differences are multi-faceted, and they encompass the lack of educational levels, time insecurity caused by unpaid care labor, safety issues, and social processes limiting the freedom of choice of women.

Importantly, gendered digital inequality is not uniform across social groups. Women's access to and use of technology is mediated by household dynamics, community expectations and cultural norms, all of which vary across regions and religious communities.

2.3 Urbanisation and Social Inequality

Urbanisation has usually been linked with better access to infrastructure, increase in incomes and exposure to modern institutions. City dwellers have a higher chance of having access to reliable electricity, mobile network connectivity and broadband access. It is based on these benefits that urbanisation has been thought to minimize social inequalities by increasing opportunities and undermining traditional limitations.

But, cities also reproduce and in certain instances aggravate the existing inequalities. The availability of infrastructure does not ensure that people have equal control over resources at home or equally engage in the life of the population. In the case of women, urban living can be accompanied by a lack of mobility, monitoring by family members, and a lack of power to make any decisions on the use of technology.

2.4 Religion as a Mediating Institution

In India, religion is one of the institutions that influence the social life. Religious communities vary in their understanding of gender roles, family patterns and admissible patterns of interaction with technology. These variations determine how women have access to digital devices, the reasons why technology is applied, and the level of freedom enjoyed by women online.

Even with this significance, religion still remains a control variable in the research on digital inequality rather than one of the main explanatory variables. This paper, through the centralization of religion, shows that it has acted as a mediating variable between urbanisation and gendered digital produces.

2.5 Intersectionality and Digital Inequality

Intersectionality is an excellent approach to studying the interaction between various axes of identity and power to realize social consequences. Instead of analysing gender, religion or sector alone, an intersectional approach acknowledges the fact that people live with these identities concurrently. This, in terms of the digital access, implies that the experiences of women in urban areas vary within the religious frames, and that the gender disparities can be experienced differently in urban and rural areas.

3. Data and Methodology

3.1 Data Source

The study uses data from the NSSO 80th Round Modular Telecom Survey, a nationally representative survey designed to capture patterns of access to and use of digital technologies in India. The survey collects detailed information on ownership and access to mobile phones and computers, types of digital activities performed, and reasons for not using. It also includes socio demographic variables such as gender, religion and sector of residence.

The unit of analysis is the individual. After excluding cases with missing values on key analytical variables, the final analytical sample consists of 136,937 individuals. The large sample size allows for detailed disaggregation by gender, religion and sector while maintaining statistical robustness.

3.2 Variable Construction

Gender

Gender is categorised as male, female and transgender. Due to the very small number of transgender respondents, most analyses focus on comparisons between men and women.Since the gender was a string variable it was converted in numeric variable using automatic recode function.

Religion Group

Religion is grouped into Hindu, Muslim, Christian, and Otherss(Combining other religion such as Sikh, Buddhism, Jainism etc.) categories for analytical clarity. Other religions are groupd because sample sizes are too small for reliable inference.

Sector

Sector distinguishes between rural and urban areas, capturing differences in infrastructure and social context.

Access to Digital Device

The variable digital_access was composed by creating a digital access index in the form of (use mobile n = 1) + 2*(use comp n =1). This variable will combine two variables that are mobile and computer access in order to make a single variable and give a value ranging between 0 to 3 that means a person no digital access, mobile access, computer access, and access to both computer and mobile.

Functional Digital Use Score (0–4)

This composite index captures engagement in functional digital activities such as education, banking, government services, and work-related tasks. Each activity contributes one point, resulting in a score ranging from 0 to 4. This measure moves beyond access to assess the quality and purpose of digital engagement.

Access to particular devices

Then acc was obtained as Any_access. This forms an uncomplicated binary marker of whether a person has any digital access or not. The variable is useful in making clean comparisons of the access differences between gender and sector using crosstabs since 0 means the respondent lacks any devices (mobile or computer) completely and 1 the respondent has at least one device (mobile or computer).

Digital Exclusion Score (0–3)

The Digital Exclusion Score combines information on lack of access to devices and reported barriers to digital use, such as affordability and lack of skills. Higher scores indicate deeper and more multidimensional exclusion.

3.3 Analytical Methods

The analysis employs cross-tabulations to examine associations between gender and digital access, supported by chi-square tests to assess statistical significance. Descriptive statistics and mean comparisons are used to analyse functional digital use and exclusion. Religion-specific analyses focus on urban areas to assess whether gender gaps persist within ostensibly advantaged contexts.

4. Results

In this section, the empirical results of the research are demonstrated with the help of the tables and charts created by the NSSO 80 th Round Modular Telecom Survey. The subsections clearly describe what the statistical tables and graphs are showing why this particular specific statistical tool is employed and how results help in answering the research question.

4.1 Gender Differences in Access to Digital Devices

Gender * Access to digital device Crosstabulation

			Access to digital device: No digital Device	Access to digital device: Digital Device	Total
Gender	male	Count	7875	61783	69658
		% within Access to digital device	37.5%	53.3%	50.9%
	female	Count	13096	53988	67084
		% within Access to digital device	62.3%	46.6%	49.0%
	transgender	Count	40	155	195
		% within Access to digital device	0.2%	0.1%	0.1%
Total		Count	21011	115926	136937

% within Access to digital device	100.0%	100.0%	100.0%

Table 1: Gender × Access to Digital Device (Crosstabulation with Counts and Percentages)

Table 1 is a cross tabulation between access to digital devices and gender. Here, the cross-tabulation is adopted due to the fact that the two variables are categorical and the aim is to analyse the distribution of access status among the categories of gender. The table summarises the numbers as percentages and absolute count, and these numbers can be interpreted in terms of magnitude and proportion.

According to the table, there were 136,937 valid cases, 21,011 of which respond that they do not have access to any digital device, and 115,926 respond that they do have access to a digital device. Women make up a disproportionately large population among the non-digital population. In particular, 13,096 of 21,011 digitally excluded people are women, which constitute 62.3 per cent of the population that is digitally excluded. Men on the contrary, represent 7875 people or 37.5 per cent. This instantly brings out the gendered aspect of digital exclusion in India.

The gender ratio is reversed among the people having access to digital devices. Females are 46.6 per cent digitally included whereas men are 53.3 per cent digitally included. This disparity suggests that men have a higher likelihood of enjoying the benefits of digital inclusion than women despite the fact that almost half of the total sample is comprised of women.

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	1780.177[a]	2	.000
Likelihood Ratio	1794.098	2	.000
Linear-by-Linear Association	1772.438	1	.000
N of Valid Cases	136937		

a. 0 cells (.0%) have expected count less than 5. The minimum expected count is 29.92.

Table 2: Chi-square test

In order to determine whether this observed association is statistically significant a chi-square test of independence is used. Pearson chi-square value equals 1780.177 and the degrees of freedom equals 2 and p-value is less than 0.001. The extent of this chi-square significance is exceedingly high, which shows that there is a strong and statistically significant relationship between gender and access, which disapproves the null hypothesis that access and gender are independent. The fact that there were no cells with a number lower than five also attests to the strength of this test.

The measures of associated symmetry reported and the chi-square test give extra information on the strength and direction of the association. The R of Pearson and the rank of Spearman are both equal to -0.114, which means that there is a negative correlation between the gender (scaled with higher values indicating a woman) and access to digital devices. Though the size of correlation is marginal, the statistical significance indicates the size of the sample and systematic disadvantage to women.

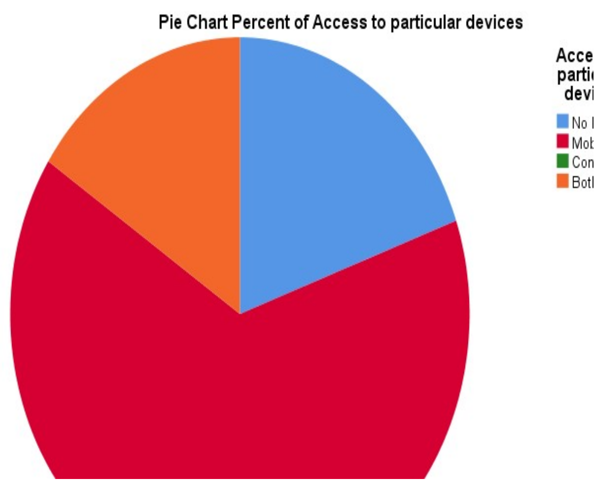

Figure 1: Percentage of Individuals With and Without Digital Access by Gender (% within Gender)

Figure 1 is a complement of Table 1 because it shows percentages of each group of gender. This visualization is essential since it changes the focus on the composition of the excluded population to the risk of exclusion of every gender. As indicated in the figure, 19.5 per cent of women indicate that they have no access to digital devices as compared to 11.3 per cent of men.

The percentage of people who lack access is even greater in transgender individuals, which is 20.5 per cent, but this population is not numerous.

Figure 1 supports the conclusion that women have a significantly greater probability of being digitally excluded than men with a visual tool, which is the clustered bar chart. This chart is put just after Table 1 since it gives an intuitive graphical outline of the same relationship that offers the gender gap more reachable to the reader.

4.2 Functional Digital Use and the Limits of Access

Although access to digital devices is one of the pre-requisites of digital participation, it does not reflect the differences in the actual use of technology. To overcome this weakness, the paper discusses the functional digital use based on a composite index.

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation	Skewness		Kurtosis	
	Statistic	Statistic	Statistic	Statistic	Statistic	Statistic	Std. Error	Statistic	Std. Error
Functional Digital Use Score (0–4)	142065	.00	4.00	1.5382	1.14912	.027	.006	-1.342	.013
Valid N (listwise)	142065								

Table 3: Descriptive Statistics of Functional Digital Use Score (0–4)

Descriptive statistics of the Functional Digital Use Score that is 0-4 is presented in Table 3. This score is used to represent the engagement in functional activities of education, banking, government services as well as work based activities. Here, descriptive statistics are employed to give a summary of the central tendency of the index and the dispersion of the index in the population.

The average Functional Digital Use Score is 1.54 and a standard deviation of 1.15. This means that people are not having two functional digital activities on average. The standard deviation is relatively large, indicating a high level of heterogeneity as there are large groups of people who utilize technology actively both functionally and those who use technology little or not at all. This has a skewed distribution with a slight skew to the left showing that the people are concentrated on the lower levels of functional use.

Report

Access to digital device		Functional Digital Use Score (0–4)	Digital Exclusion Score (0–3)
No digital Device	Mean	.0000	3.0000
	N	21011	21011
	Std. Deviation	.00000	.00000
Digital Device	Mean	1.8851	1.7687
	N	115926	115926
	Std. Deviation	.98203	.42167
Total	Mean	1.5958	1.9576
	N	136937	136937
	Std. Deviation	1.13049	.58946

Table 4: Mean Functional Digital Use Score by Access to Digital Device

Table 4 will compare the mean scores of functional use in those who have and do not have access to digital devices. The mean analysis is applied in this comparison since the Functional Digital Use Score is a numerical index, and the aim is to evaluate the differences in the average engagement between the categories of access.

The findings show a dramatic difference. Those who do not have access to digital tools are those who have a mean functional use score of 0.00 meaning that they are totally not involved in functional digital activities. On the contrary, those who can access digital devices are at 1.89 with a standard deviation of 0.98. At this difference, access to functional digital participation becomes central.

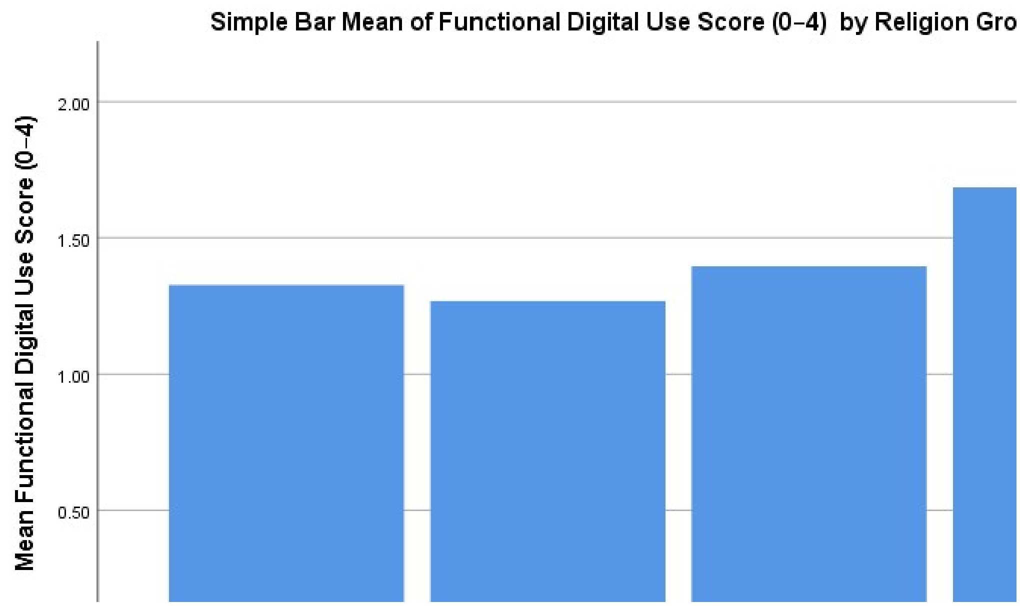

Figure 2: Mean Functional Digital Use Score by Access Status

Table 3 was visualized in Figure 2. With the help of a bar chart, the figure will be able to see that the access to the digital devices is closely related to the increased functional use. Yet, the fact that the mean point in digitally included people is not above 2 also proves that access per se does not presuppose widespread and diversified usage. This figure is inserted just below Table 4 to enhance the comparison of numerical data in the table.

4.3 Digital Exclusion as a Multidimensional Phenomenon

Digital exclusion extends beyond the absence of devices to include barriers such as lack of skills, affordability, and social constraints. To capture this complexity, the study uses a Digital Exclusion Score.

Digital Exclusion Score (0–3)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1.00	26815	18.9	19.6	19.6
	2.00	89111	62.7	65.1	84.7
	3.00	21011	14.8	15.3	100.0
	Total	136937	96.4	100.0	
Missing	System	5128	3.6		

Total	142065	100.0		

Table 5: Distribution of Digital Exclusion Score (0–3)

Table 5 presents the frequency distribution of the Digital Exclusion Score. The table shows that 18.9 per cent of individuals have an exclusion score of 1, indicating limited exclusion. A substantial majority, 62.7 per cent, have a score of 2, suggesting partial exclusion due to one or more barriers. Finally, 14.8 per cent have the highest exclusion score of 3, indicating deep and multidimensional exclusion.

The fact that nearly two-thirds of individuals fall into the middle category underscores that digital exclusion is not an all-or-nothing phenomenon. Many individuals experience partial inclusion, where access exists but effective use is constrained.

4.4 Religion-Sensitive Urban Gender Inequality

The core objective of this study is to assess whether urbanisation reduces gender inequality uniformly across religious groups. This subsection focuses exclusively on urban women and examines how exclusion varies by religion.

Religion Group * Urban_gender_gap Crosstabulation

			Urban_gender_gap		
			.00	1.00	Total
Religion Group	1.00	Count	49368	11564	60932
		% within Religion Group	81.0%	19.0%	100.0%
	2.00	Count	3468	1241	4709
		% within Religion Group	73.6%	26.4%	100.0%
	3.00	Count	577	166	743
		% within Religion Group	77.7%	22.3%	100.0%
	4.00	Count	575	125	700
		% within Religion Group	82.1%	17.9%	100.0%
Total		Count	53988	13096	67084
		% within Religion Group	80.5%	19.5%	100.0%

Table 6: Religion Group × Urban Gender Gap Crosstabulation

Table 6 presents a cross-tabulation between religion group and an indicator identifying urban women without digital access. The use of cross-tabulation is appropriate here because both variables are categorical, and the aim is to compare exclusion rates across religious groups.

The results reveal marked variation. Among Hindu women in urban areas, 19.0 per cent are digitally excluded. In contrast, the exclusion rate among Muslim women is substantially higher at 26.4 per cent. Christian women also exhibit elevated exclusion at 22.3 per cent, while Sikh women show relatively lower exclusion at 17.9 per cent. These differences demonstrate that religion significantly mediates the relationship between urban residence and gendered digital access.

5. Discussion

The results of this research are strong indicators that urbanisation, being related to an increase in overall rates of digital access, does not necessarily decrease gender inequality. The analysis above indicates clearly that women are still outnumbered with regard to the ownership of digital devices as compared to men. This trend continues to recur even in cities, which suggests that the spatial closeness of digital infrastructure is not enough to break the established social inequalities.

The first lesson that can be made as a result of the analysis is the difference between the availability of infrastructure and social empowerment. As much as urban environments are more accessible in terms of accessibility of digital networks, devices, and services, the accessibility of such facilities to women is limited. The findings indicate that gender relations of power in families and society still define the question of who owns digital devices and their applications. In turn, urbanisation opens the possibilities on digital inclusion, yet the potential does not imply the fair access and usage results.

The findings also prove that gender inequality in online access does not cut across social groups equally. The variations found across the religious groups are clear indicators that culture and social expectations are of paramount importance in mediating the female digital engagement. Women belonging to certain religious groups have a greater degree of digital exclusion even in vicinities with a relatively similar degree of infrastructure. This implies that urbanisation does not exterminate cultural and normative boundaries but that it plays complex interaction with cultural and normative boundaries.

The other valuable aspect of the work is the shift towards a digital access that is more binary in nature. The functional analysis of digital use indicates that device access does not always imply meaningful and diversified use of digital technologies. Numerous digitally incorporated people do not use more than a narrow spectrum of activities, whilst others are not functional users of digital devices like education, financial services, or communication to government websites. This explains why it is important to investigate the quality and purpose of digital use than ownership or access only.

The analysis of the digital exclusion also supports the idea of the multidimensionality of digital inequality. The fact is that exclusion tends to work in stratified ways, uniting absence of access with obstacles, including the lack of ability, the inability to afford and social restrictions. The percentage of people who are partially excluded is significant, which indicates that digital disadvantage is not a divide but it exists in a continuum between users and non-users. This questions excessive simplistic accounts of digital inclusion that emphasize measures of connectivity.

In combined form, the results highlight the significance of tying the results on an individual level, in terms of digital outcomes, to the larger social systems. The interactions of urbanisation, gender norms and religious identities influence access to the digital in a manner that cannot be explained by looking at one of these elements individually. Instead of being a neutral process, urban development replicates and reorders existing practices of social inequalities. These interactions are critical in explaining why technological expansion does not necessarily result in social inclusion.

6. Conclusion and Policy Implications

This paper aimed at investigating the impact of urbanisation on gender difference in access to digital in India using a religion-sensitive approach. Based on nationally representative statistics, the analysis will indicate that although urbanisation does lead to a significant improvement in the general level of digital access, gender-based differences cannot be eradicated. Women are still overrepresented among the digitally excluded people, and the level of exclusion differs depending on religious communities in urban areas.

These findings show clearly that in India digital inequality is not only a by-product of inequality in infrastructure but also entrenched in social relations. Cultural expectations, gender norms, and religious contexts remain in control of beneficiaries of the digital technologies. Consequently, the issue of space transformation cannot resolve deep-rooted inequality issues. Urbanisation can also lower some structural barriers but it does not necessarily change social norms of access and use.

Policy-wise, these results imply that the policies of digital inclusion should cease their confined trend of focusing on making things more connected and accessible. Although it requires infrastructure development, it is not enough to provide equitable digital participation. There should be policies which clearly capture the social and cultural constraints that restrain the access and effective use of digital technologies by women.

Sensitive digital literacy programs are needed on gender basis, especially in cities where there is accessibility but scanty use. Those initiatives must also be aimed at confidence, independence, and awareness of the opportunities of digital engagement rather than only technical skills. Economic barriers can also be reduced by subsidised devices and inexpensive data plans as this will limit the number of women who experience a disproportionate impact.

Simultaneously, the interventions of a policy should be culturally responsive. The challenge with a universal approach to digital inclusion is that it will ignore the unique limitations women will experience in various religious groups. It is possible that community-based initiatives involving families, institutional participation, and social leaders are more effective in solving restrictive norms and enabling women to participate in digital.

The paper also indicates some future research opportunities. The longitudinal data may be employed to determine whether the gendered digital inequalities reduce or continue to decline as urbanisation progresses. Qualitative research may offer more information on the daily negotiations and power relations which drive access to technology by women. The intersections with other dimensions of inequalities: education, caste, and employment may also be examined in further work.

To summarise, this paper has shown that digital inequality in India is more a social than technological challenge. Digital inclusion is a valuable opportunity produced by urbanisation, and unless these opportunities are focused on an underlying social structure that consolidates access and use, they will be unevenly distributed. Holism that combines technological growth with social and cultural interventions is the key to successful and relevant digital transformation.

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